METRIC

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SUPERSEDING
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(See 6.5)

MILITARY SPECIFICATION

COLORS, TINTING, FOR INTERIOR NONFLAMING (DRY) PAINTS (METRIC)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers tinting colors to be used by mixing with enamel, nonflaming (dry), chlorinated alkyd resin, soft white semigloss Formula No. 124 (type V of MIL-E-17970) to produce interior finish paints for living spaces on ships. Products covered by this specification are suitable for use where air pollution regulations apply.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
TT-T-291 - Thinner-Paint, Volatile Spirits, Petroleum Spirits.
PPP-P-1892 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.

MILITARY
MIL-A-15206 - Aluminum Stearate, Technical.
MIL-E-17970 - Enamel, Nonflaming (Dry), Chlorinated Alkyd Resin, Soft White, Semigloss, Formula No. 124/58.
MIL-R-21417 - Resin, Chlorinated Alkyd, Solution.

STANDARDS

FEDERAL
FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
FED-STD-313 - Material Safety Data Sheets, Preparation and the Submission of.
FED-STD-595 - Colors.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 209 - Lampblack, Spec. for.
D 211 - Chrome Yellow and Chrome Orange, Spec. for.
D 261 - Iron Blue, Spec, for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8010

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) (Continued)

D 562 - Consistency of Paints Using the Stormer Viscosimeter, Test for.
D 600 - Liquid Paint Driers, Spec. for.
D 656 - Pure Toluidine Red Toner, Spec. for.
D 1210 - Fineness of Dispersion of Pigment-Vehicle Systems, Test for.
D 1296 - Odor of Volatile Solvents and Diluents, Test for.
D 1364 - Water in Volatile Solvents (Fischer Reagent Titration Method), Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Toxicity. The material shall have no adverse effect on the health of personnel when used for its intended purpose (see 4.5). Questions pertinent to this effect shall be referred by the procuring activity to the appropriate service medical department which will act as advisor to the procuring activity.

3.2 Formula.

3.2.1 The tinting colors shall consist of ingredients conforming to the applicable specifications in the proportions shown in table I, except that the pigment proportions may be varied slightly to conform to the color requirement. The exact formula used shall be shown on the forms required by the inspection procedure of method 1031 of FED-STD-141 (see 4.2).

TABLE I. Tinting color formulas.

Ingredients	kg (Pounds) per 100 gallons 1/									
	Rose Wood	Beach Sand	Yellow Gray	Green Gray	Pearl Gray	Sun Glow	Clipper Blue	Pastel Blue	Pastel Green	Blkhd Gray
Chrome yellow (ASTM D 211)	145.455 (320)	128.182 (282)	91.818 (202)	55.455 (122)	15.0 (33)	284.091 (625)	68.182 (150)	---	45.455 (100)	136.364 (300)
Toluidine red (ASTM D 656)	28.182 (62)	15.909 (35)	6.0 (13.2)	2.636 (5.8)	---	4.091 (9)	---	1.091 (2.4)	---	7.273 (16)
Lampblack (ASTM D 209)	1.182 (2.6)	1.409 (3.1)	4.0 (8.8)	8.682 (19.1)	9.545 (21)	0.182 (0.4)	5.409 (11.9)	4.727 (10.4)	---	3.182 (7)
Iron blue (ASTM D 261)	---	---	---	---	---	---	5.909 (13.0)	6.591 (14.5)	5.909 (13.0)	---
Magnesium silicate 2/	52.273 (115)	81.818 (180)	110.455 (243)	126.818 (279)	136.364 (300)	28.182 (62)	123.182 (271)	139.091 (306)	136.364 (300)	81.818 (180)
Aluminum stearate (MIL-A-15206)	4.545 (10)	4.545 (10)	4.545 (10)	4.545 (10)	4.545 (10)	4.545 (10)	4.545 (10)	4.545 (10)	4.545 (10)	4.545 (10)
Chlorinated alkyd resin (MIL-R-21417)	181.818 (400)	168.182 (370)	163.636 (360)	156.818 (345)	156.818 (345)	197.727 (435)	163.636 (360)	156.818 (345)	156.818 (345)	172.727 (380)
Petroleum spirits (type II, grade A of TT-T-291)	107.273 (236)	119.091 (262)	124.091 (273)	130.0 (286)	134.091 (295)	95.455 (210)	124.091 (273)	134.091 (295)	134.091 (295)	118.182 (260)
Lead naphthenate drier (class B of ASTM D 600)	2.273 (5.0)	2.136 (4.7)	2.045 (4.5)	2.0 (4.4)	1.955 (4.3)	2.5 (5.5)	2.045 (4.5)	1.955 (4.3)	2.045 (4.5)	2.136 (4.7)
Cobalt naphthenate drier (class B of ASTM D 600)	0.955 (2.1)	0.864 (1.9)	0.864 (1.9)	0.818 (1.8)	0.818 (1.8)	1.0 (2.2)	0.864 (1.9)	0.818 (1.8)	0.864 (1.9)	0.955 (2.1)
Antiskinning agent	0.455 (1.0)	0.455 (1.0)	0.455 (1.0)	0.455 (1.0)	0.455 (1.0)	0.455 (1.0)	0.455 (1.0)	0.455 (1.0)	0.455 (1.0)	0.455 (1.0)

See footnotes at top of next page.

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1/ The formula is given slightly in excess of 100 gallons to allow for normal manufacturing loss.

2/ The magnesium silicate shall be a micronized grade equal to "S.F. Magnesium Silicate" marketed by Whittaker-Clark and Daniels.

3/ Antiskinning agent shall be a volatile oxime type equal to methyl ethyl ketoxime.

3.3 Manufacture. The component raw materials shall be mixed and ground as required to produce a tinting color which is uniform, easily dispersed in the white base (Formula No. 124), free from grit, free from color separation, and in complete conformity to the requirements of this specification.

3.4 Quantitative requirements.

3.4.1 The tinting colors, prepared as specified (see table I), shall conform to table II when tested as specified (see 4.4).

TABLE II. Quantitative requirements.

Characteristic	Minimum	Maximum
Water, percent by weight of tinting color	---	0.5
Coarse particles and skins (as residue retained on standard No. 325 sieve) percent by weight of tinting color	---	0.5
Consistency, Krebs-Stormer shearing rate 200 RPM		
grams	125	175
equivalent Krebs units	67	77
Fineness of grind	6	---

3.4.2 Volatile portion. The volatile portion of the tinting colors shall conform to the following requirements by volume:

(a) A combination of hydrocarbons, alcohols, aldehydes, ethers, esters, or ketones having an olefinic or cycloolefinic type of unsaturation except perchloroethylene: 5 percent maximum.
(b) A combination of aromatic compounds with eight or more carbon atoms to the molecule except ethylbenzene, methyl benzoate, and phenyl acetate: 8 percent maximum.
(c) A combination of ethylbenzene, ketones having branched hydrocarbon structures, trichloroethylene, or toluene: 20 percent maximum.

3.5 Qualitative requirements.

3.5.1 Odor. The odor shall be characteristic of the volatiles permitted in 3.4.2.

3.5.2 Color. When the contents of a 1/2-pint can of a tinting color is mixed as directed in 4.4.2.2 with 1 gallon of soft white chlorinated alkyd resin enamel (Formula No. 124), the dry film of the resultant paint shall acceptably match the applicable standard color of FED-STD-595 as specified (see Table III).

TABLE III. Color numbers of paints made from the tinting pigments.

Color	Rose Wood	Beach Sand	Yellow Gray	Green Gray	Pearl Gray	Sun Glow	Clipper Blue	Pastel Blue	Pastel Green	Bulkhead Gray
FED-STD-595 Color No.	22519	22563	26400	26496	26493	23697	24516	25526	24585	26307

3.5.3 Compatibility. There shall be no evidence of incompatibility of any the ingredients of the tinting color or of the resultant paint when mixed as described in 4.4.2.2 .

3.5.4 Condition in container. For at least 1 year after date of manufacture, the tinting color, prepared as specified (see table I), shall be capable of being readily broken up with a paddle to a smooth, uniform consistency, shall not exceed 90 Krebs units in viscosity, and shall not liver, curdle, gel, or show any other objectionable properties.

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3.6 Material safety data sheet. The procuring activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS is DD Form 1813 found in and part of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Quality conformance inspection shall be provided in accordance with method 1031 of FED-STD-141 as hereinafter supplemented. Submission of forms in accordance with method 1031 shall cover complete tests other than the 1-year condition in container test. The condition in container requirement shall apply for 1 year after manufacture regardless of other testing or prior acceptance of the material. Inspection shall particularly determine precision of fill.

4.2.1 Ingredient materials. When requested by the controlling authority, 1 pint of the ingredients in the formula in table I shall be provided for test purposes.

4.3 Acceptance. Acceptance of the tinting color shall be based upon conformance of the tinting color to the requirements of this specification. Failure of any test shall be cause for rejection of the lot.

4.4 Test procedures.

4.4.1 Ingredient materials. All ingredient materials submitted shall be tested to determine compliance with the applicable specifications listed in table I.

4.4.2 Finished tinting color.

4.4.2.1 The following tests shall be conducted in accordance with the methods specified in table IV:

TABLE IV. Test methods.

Tests	Applicable method in FED-STD-141	Applicable ASTM test method
Water	---	D 1364
Coarse particles and skins	4092	---
Consistency	---	D 562
Fineness of grind	---	D 1210
Odor	---	D 1296

4.4.2.2 Color. The color shall be determined as follows: Add the entire contents of the 1/2-pint can of tinting color to 1 gallon of soft white chlorinated alkyd resin paint (Formula No. 124) and mix thoroughly. (A suggested procedure is as follows: After draining the 1/2-pint can of tinting color into the soft white paint with constant stirring, partly refill the 1/2-pint can with some of the mixed paint, stir, and return to the gallon of mixed paint. Stir until completely uniform.) Prepare a dry film of the mixed paint by using a film applicator (doctor blade) to draw down a film on a planar piece of opaque white glass at a wet film thickness of approximately 0.0076 centimeter (cm). After a 48-hour air-drying period, compare the color of the film with the pertinent standard color card (see table III) in accordance with method 4250 of FED-STD-141. If doubt exists as to the satisfaction of the match, the color shall show no more than two color units difference from the standard when tested in accordance with method 6123 of FED-STD-141.

4.5 Toxicity. A manufacturer of material shall disclose the formulation of his product to the Navy Bureau of Medicine and Surgery, Navy Department, Washington, DC 20372. The disclosure of proprietary information, which shall be held in confidence by the Bureau of Medicine and Surgery, shall include: the name, formula, and approximate percentage by weight and volume of each ingredient in the product; the results of any toxicological testing of the product; identification of its pyrolysis products; and any such other information as may be

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eeded to permit an accurate appraisal of any toxicity problem associated with the handling, torage, application, use, or disposal of the material.

4.6 Inspection of Preparation for Delivery. Sample packages and packs and the inspection of packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements.)

5.1 Packaging, packing and marking. The tinting colors shall be packaged level A or C, packed level A, B or C, and marked, as specified (see 6.2), in accordance with PPP-P-1892. The tinting colors shall be furnished in 1/2-pint cans unless otherwise specified (see 6.2).

5.1.1 Special marking. In addition to other markings required on the containers, there shall be the following statements: "The volatile content of the material in this container is not photochemically reactive as defined by rule 102 of the South Coast Air Quality Management District." (See 6.4.)

"CAUTION: These colors contain toxic lead compounds. Do not use on interior surfaces, porches, or exterior surfaces of any items of furniture, machinery, or appliance to which children may be commonly exposed." (See 6.1 and also Public Law 91-695.)

6. NOTES

6.1 Intended use. The tinting colors are intended for use in tinting enamel, nonflaming (dry), chlorinated alkyd resin, soft white semigloss Formula No. 124 (type V of MIL-E-17970) to produce interior finish paints for living spaces on ships duplicating the enamels covered by MIL-E-17970. The tinting colors permit maintaining a wider selection of colors under conditions of limited storage space and inventory cost than possible with pretinted paints.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Color of tint color required (see 1.1).
(b) Levels of packaging and packing required (see 5.1).
(d) Size of container (see 5.1).
(e) Special marking required (see 5.1.1).

6.3 Tinting colors should be purchased by volume, the unit being a 1/2 U.S. pint at 15.6°C.

6.4 Volatile content. Although the container marking specifically refers to the South Coast Air Quality Management District, the paint may be used anywhere else paint complying with 3.4.2 is allowed. This includes other air pollution control districts or similar areas controlling this emission of solvents into the atmosphere. Information regarding Los Angeles County Air Pollution Rule 102, 442, and 443 may be obtained from: South Coast Air Quality Management District, Metropolitan Zone, 434 South San Pedro Street, Los Angeles, California 90013.

6.5 Changes from previous issue. The symbol "0" is not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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